MPI Parallelisation of Molecular Dynamics Algorithms

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Abstract

A C++17/MPI molecular dynamics solver is implemented and validated using three numerical integrators—Forward Euler, fourth-order Runge–Kutta (RK4), and Velocity-Verlet—applied to a harmonic oscillator verification suite and a Lennard-Jones argon fluid ($N = 864$, $\rho = 0.8\sigma^{-3}$, $T = 94.4\,\mathrm{K}$) following Rahman 7. Harmonic oscillator convergence tests confirm global truncation orders of $\mathcal{O}(\Delta t)$, $\mathcal{O}(\Delta t^2)$, and $\mathcal{O}(\Delta t^4)$ (fitted slopes 1.05, 2.00, 3.94); RK4 error saturates at the IEEE 754 double-precision floor ($\approx 3.6 \times 10^{-15}$). For the argon fluid (100-step NVE run), Velocity-Verlet achieves bounded energy conservation ($\max|\Delta E/E_0| = 0.082\%$) while Forward Euler diverges catastrophically ($\max|\Delta E/E_0| = 127.6\%$, temperature runaway to $\sim 400\,\mathrm{K}$), confirming that symplecticity is essential for stable MD. Radial distribution function peaks at $r/\sigma = 1.09$ ($g = 2.84$) and $r/\sigma = 2.07$ ($g = 1.25$) compare favourably with Rahman's published liquid-structure data. Strong scaling at fixed $N = 2048$ yields speedup $24.1\times$ at $P = 32$ (efficiency 75%; Amdahl $f \approx 0.010$); size scaling at fixed $P = 16$ gives wall time $\propto N^{1.85}$, consistent with $\mathcal{O}(N^2)$ direct force evaluation.

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1 Introduction

Molecular dynamics (MD) simulates the time evolution of interacting particle systems by numerically integrating Newton's equations of motion. Since Rahman 7's landmark study of liquid argon, and the earlier hard-sphere simulations of Alder & Wainwright [1], MD has become central to computational chemistry, materials science, and biophysics, yielding thermodynamic properties, structural correlations, and transport coefficients inaccessible to analytic theory [2, 4]. Applications span protein folding, radiation damage in nuclear materials, phase-transition kinetics, and the design of pharmaceutical binding sites.

The computational bottleneck is pairwise force evaluation, scaling as $\mathcal{O}(N^2)$ for direct summation over N particles. Physically meaningful simulations require $N \sim 10^3$–10^6 and 10^6–10^9 timesteps; this is intractable without parallelism. Distributing the force computation across P cores via the Message Passing Interface (MPI) reduces the per-rank cost to $\mathcal{O}(N^2/P)$, but practical efficiency depends on the communication overhead required to synchronise particle data between ranks.

Modern CPU architectures expose parallelism at multiple levels: vector units within a core, multi-core sockets, and distributed-memory clusters connected by a communication fabric. For distributed-memory problems, MPI remains the standard approach because it gives explicit control over data ownership and communication, at the cost of making the programmer responsible for where the data reside and when they are exchanged.

Integrator choice is equally consequential. Long-time stability demands *symplectic* integrators that preserve the symplectic structure of Hamiltonian mechanics: non-symplectic schemes exhibit systematic energy drift regardless of formal truncation order, corrupting thermodynamic averages over extended simulations. In many-particle systems, exact long-time trajectory prediction is in any case impossible because of Lyapunov instability; the practical goal of MD is therefore not to recover one exact future trajectory, but to sample the correct thermodynamic ensemble and preserve the correct short-time dynamical behaviour as accurately as possible.

This report presents a C++17/MPI solver implementing Forward Euler, RK4, and Velocity-Verlet. The integrators are assessed for convergence order, symplecticity, and energy conservation on the harmonic oscillator and on a Lennard-Jones (LJ) argon fluid following Rahman 7. The particle-decomposition parallelisation is benchmarked for strong scaling (up to $P = 32$) and size scaling (up to $N = 2048$), and related to expected theoretical properties.

2 Methodology

2.1 Numerical Integrators

We consider $m\ddot{\mathbf{x}}_i = \mathbf{F}_i(\{\mathbf{x}\})$, reformulated as the first-order system $\mathbf{y} = (\mathbf{x}, \mathbf{v})^\top$, $\dot{\mathbf{y}} = (\mathbf{v}, \mathbf{F}/m)^\top$, with acceleration $\mathbf{a}_n = \mathbf{F}_n/m$.

2.1.1 Forward Euler

Plain forward Euler advances position and velocity by simultaneous first-order Taylor truncations:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{v}_n\,\Delta t, \tag{1}$$
$$\mathbf{v}_{n+1} = \mathbf{v}_n + \mathbf{a}_n\,\Delta t. \tag{2}$$

Both updates are first-order; the global truncation error is $\mathcal{O}(\Delta t)$. The scheme is not symplectic: it does not preserve the symplectic two-form of Hamiltonian mechanics, so phase-space volume is not conserved and total energy drifts monotonically.

2.1.2 Fourth-Order Runge–Kutta

RK4 evaluates four stage derivatives of the first-order system:

$$\mathbf{k}_j = \Delta t\,\mathbf{f}(\mathbf{y}_n + c_j\,\mathbf{k}_{j-1}), \quad j = 1, \ldots, 4, \tag{3}$$
$$\mathbf{y}_{n+1} = \mathbf{y}_n + \tfrac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4),$$

with coefficients $(c_1, c_2, c_3, c_4) = (0, \frac{1}{2}, \frac{1}{2}, 1)$. Global error is $\mathcal{O}(\Delta t^4)$. RK4 is not symplectic. Crucially, each intermediate stage requires force evaluation at an intermediate particle configuration; for the LJ system this entails a separate `MPI_Allgatherv` per stage—four per step—making RK4 prohibitively expensive for large-scale interacting systems.

2.1.3 Velocity-Verlet

The Velocity-Verlet algorithm [8, 10] implements a half-kick / drift / force / half-kick sequence:

$$\mathbf{v}_{n+1/2} = \mathbf{v}_n + \tfrac{\Delta t}{2}\,\mathbf{a}_n, \tag{4}$$
$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\,\mathbf{v}_{n+1/2}, \tag{5}$$
$$\mathbf{a}_{n+1} = \mathbf{F}(\mathbf{x}_{n+1})/m, \tag{6}$$
$$\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \tfrac{\Delta t}{2}\,\mathbf{a}_{n+1}. \tag{7}$$

The global truncation error in trajectory coordinates is $\mathcal{O}(\Delta t^2)$. The underlying Verlet construction arises from cancellation of odd-order Taylor terms when combining forward and reverse expansions, giving a local truncation error $\mathcal{O}(\Delta t^4)$ per step; the $\mathcal{O}(\Delta t^2)$ global rate reflects accumulation over $T/\Delta t$ steps and should not be conflated with the local-order result. The scheme is *symplectic* and time-reversible: it conserves a shadow Hamiltonian $\tilde{H} = H + \mathcal{O}(\Delta t^2)$, so total energy oscillates with bounded amplitude indefinitely rather than drifting [6]. This matters particularly in chaotic many-body systems, where trajectories diverge exponentially under small perturbations: the correct requirement is therefore not exact trajectory matching at long times, but accurate ensemble sampling and faithful preservation of invariants over long runs. Only one force evaluation per step is required, making Velocity-Verlet the standard choice for production MD.

2.2 Lennard-Jones Potential and Force Computation

The LJ pair potential and force on particle i are

$$V(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right], \tag{8}$$

$$\mathbf{a}_i = \frac{24\varepsilon}{m}\sum_{j\neq i}\frac{\mathbf{x}_i - \mathbf{x}_j}{r_{ij}^2}\left[2\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right], \tag{9}$$

with argon parameters $\varepsilon/k_\mathrm{B} = 120\,\mathrm{K}$, $\sigma = 3.4\,\text{Å}$, $m = 66.904\times10^{-27}\,\mathrm{kg}$ [7]. A hard cutoff is applied at $r_c = 2.25\sigma$; the geometric constraint $r_c < L/2 \approx 5.115\sigma$ (for $N = 864$, $L = 10.229\sigma$) ensures validity of the minimum image convention $\delta x \leftarrow \delta x - L\,\mathrm{round}(\delta x/L)$. Potential energy is accumulated unconditionally over all $j \neq i$ pairs and multiplied by 0.5 post-loop (avoiding a branch in the $\mathcal{O}(N^2)$ hot loop). Newton's third law is not exploited (see §2.3). The force kernel uses only integer powers: $r^{-2} \to r^{-6} \to r^{-12}$, with all constants precomputed at compile time.

Initial conditions follow Rahman: an FCC lattice ($N = 4k^3$, $k = 6$) with random perturbations $\pm0.01\sigma$, and Maxwell–Boltzmann velocities via the Box–Muller transform. Temperature is computed from equipartition using $N_\mathrm{dof} = 3(N-1)$ degrees of freedom after removing centre-of-mass drift. Box length scales as $L = 10.229\sigma\,(N/864)^{1/3}$ to maintain constant number density.

2.3 Parallelisation Strategy

Particle decomposition. Rank r owns a fixed, contiguous block of particles $[\mathrm{offset}_r, \mathrm{offset}_r + N_r^\mathrm{loc})$, where $N_r^\mathrm{loc} = \lfloor N/P \rfloor + (r < N \bmod P\,?\,1:0)$. Ownership is determined at startup and fixed throughout the simulation.

Data layout. Particle state is stored in flat interleaved `std::vector<double>` arrays of size $3N_r^\mathrm{loc}$, indexed as `arr[3*i+d]`. This layout provides sequential cache access in force loops and maps directly to `MPI_Allgatherv` buffers without packing. Velocities and forces are never communicated.

MPI API usage. Five collective operations are used:

1. `MPI_Bcast`: rank 0 generates all initial conditions (`std::mt19937_64`, seed 42) and broadcasts the complete $3N$-double position and velocity arrays before the time loop.
2. `MPI_Allgatherv`: the sole collective in the time-stepping loop. After updating and wrapping local positions, each rank contributes its $3N_r^\mathrm{loc}$ doubles to reconstruct the global position array on every rank, *before* force evaluation. Correct sequencing—wrap → allgather → force compute—is critical: forces require globally consistent positions. For the non-interacting harmonic oscillator, this call is bypassed entirely.
3. `MPI_Reduce(MPI_SUM)`: local kinetic and potential energies are summed to rank 0 for output.
4. `MPI_Reduce(MPI_MAX)`: maximum wall time across all ranks is reported; separate reduces capture communication statistics.
5. `MPI_Barrier` / `MPI_Wtime`: synchronise ranks before timing; accumulate `MPI_Allgatherv` wall time when the timing flag is set.

Newton's Third Law trade-off. Each rank evaluates forces on its local particles against all N particles, giving $\mathcal{O}(N^2/P)$ rather than $\mathcal{O}(N^2/2P)$ cost. This avoids the complex partial-force exchange communication required by third-law optimisation, keeping a single clean `MPI_Allgatherv` per step (four for RK4). For RK4, the allgather is called once per intermediate stage.

Equilibration and reproducibility. A 50-step startup phase applies per-step velocity rescaling ($\lambda = \sqrt{T_\mathrm{target}/T_\mathrm{meas}}$), broadcast via `MPI_Bcast`, followed by a final exact rescale at the startup-to-production boundary. The 100-step production run is clean NVE. All timing runs use identical initial conditions (fixed seed) and are performed without file output, as required by the assignment brief.

Communication-cost perspective. The present particle-decomposition algorithm is deliberately simple and transparent, but it exposes the standard distributed-memory trade-off: computation is reduced by splitting the force loop across ranks, while communication incurs latency and bandwidth costs every time global particle positions are reconstructed. For strong scaling, the amount of work per rank decreases as P increases, so communication occupies a larger fraction of the runtime even if its absolute cost remains modest.

2.4 Validation Methodology

Parallel correctness is verified by confirming that energy trajectories agree between $P = 1$ and $P = 2$ runs to bitwise precision. For $P > 2$, floating-point non-associativity in `MPI_Reduce` tree reductions introduces differences at $\sim10^{-12}$–10^{-14} relative tolerance, consistent with IEEE 754 arithmetic. This is expected whenever the order of floating-point summation changes between serial and parallel executions.

3 Results 1: Harmonic Oscillator Verification

The harmonic oscillator $\ddot{x} = -\omega^2 x$ ($\omega = 1$, $x_0 = 1$, $\dot{x}_0 = 0$) has exact solution $x(t) = \cos t$, providing a stringent benchmark against which convergence and energy behaviour can be assessed analytically.

Trajectories and phase space

Figure 1 shows position, velocity, and phase-space trajectories at $\Delta t = 0.01$ over $T = 10$. All three integrators track the exact solution closely at this resolution; the insets reveal that Euler accumulates a visible phase error near $t = T$. In phase space (panel c), Euler traces an outward spiral consistent with non-symplectic energy growth and violation of Liouville's theorem; both Velocity-Verlet and RK4 remain on the exact unit circle.

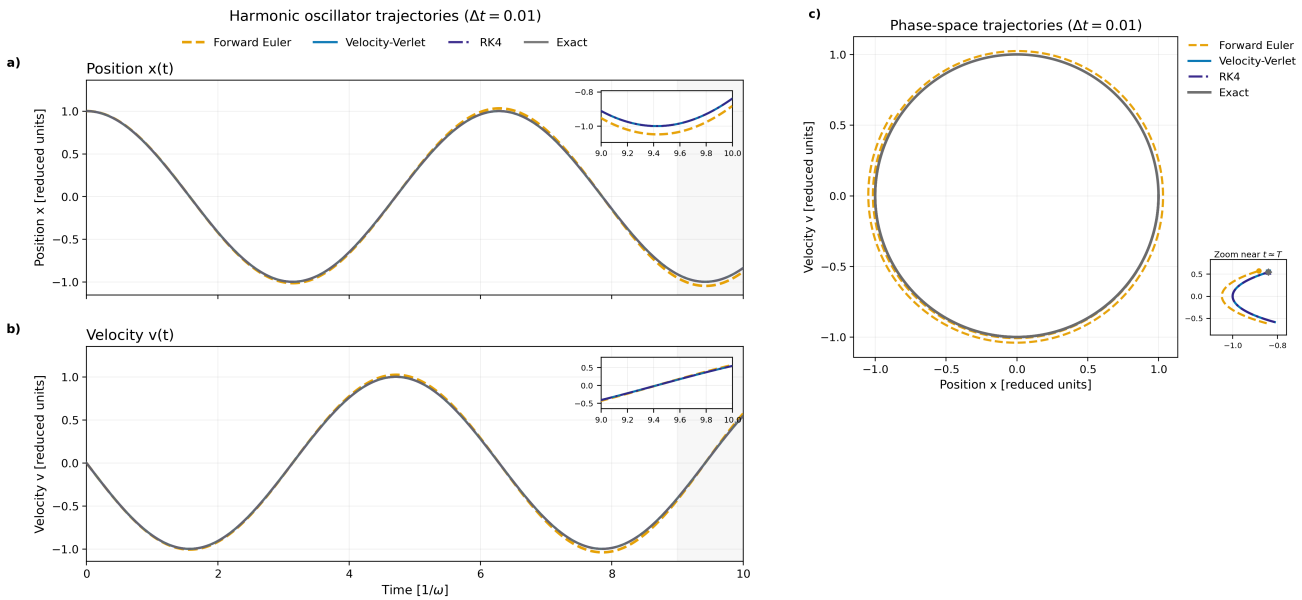

Figure 1. Harmonic oscillator trajectories at $\Delta t = 0.01$, $\omega = 1$, $T = 10$; exact solution in grey. **(a)** Position $x(t)$ and **(b)** velocity $v(t)$; insets zoom near $t \approx T$. Velocity-Verlet (blue) and RK4 (dark blue dash-dot) overlay the exact solution; Euler (orange dashed) shows visible phase deviation. **(c)** Phase-space portrait. Euler spirals outward (non-symplectic); Velocity-Verlet traces a near-exact closed orbit (symplectic, Liouville-preserving); RK4 is indistinguishable from the exact circle.

Table 1. Global position error $|x_{\text{num}}(T{=}10) - x_{\text{exact}}|$ and fitted convergence slopes.

Δt	Euler	Vel.-Verlet	RK4
10^0	7.0×10^0	3.4×10^{-1}	2.2×10^{-2}
10^{-1}	2.3×10^{-1}	2.3×10^{-3}	3.9×10^{-6}
10^{-2}	2.0×10^{-2}	2.3×10^{-5}	4.5×10^{-10}
10^{-3}	2.0×10^{-3}	2.3×10^{-7}	5.3×10^{-14}
5×10^{-4}	9.8×10^{-4}	5.7×10^{-8}	3.6×10^{-15}†
Fitted slope	1.05	2.00	3.94

†IEEE 754 double-precision floor.

Small vs. large timestep

Figure 2 illustrates the effect of timestep size. At $\Delta t = 0.5$, Euler is numerically unstable (amplitude growth to $|x| \approx 8$, spiralling outward in phase space); Velocity-Verlet remains bounded even at this coarse resolution—a key property of symplectic integrators, which preserve qualitative dynamics even when accuracy is poor; RK4 stays close to the exact trajectory in both time and phase space.

Convergence order

Figure 3 presents log–log convergence curves for the endpoint position error $|x_{\text{num}}(T{=}10) - x_{\text{exact}}(T{=}10)|$ and the RMS phase-space error, fitted over the asymptotic regime $\Delta t \leq 0.1$. Table 1 reports representative errors and fitted slopes.

Fitted slopes of 1.05, 2.00, and 3.94 confirm the expected first-, second-, and fourth-order convergence. At $\Delta t = 10^{-2}$, Velocity-Verlet is three orders of magnitude more accurate than Euler, and RK4 a further five orders more accurate than Verlet. RK4 saturates at $\approx 3.6 \times 10^{-15}$ for $\Delta t = 5 \times 10^{-4}$, at the IEEE 754 double-precision floor.

The $\mathcal{O}(\Delta t^2)$ global convergence of Velocity-Verlet arises from accumulation of steps whose local truncation error is formally $\mathcal{O}(\Delta t^4)$, due to cancellation of odd-order terms in the symplectic derivation; this local accuracy is not equivalent to fourth-order global convergence.

Energy conservation

Figure 4 shows the relative energy deviation $(E - E_0)/|E_0|$ over $T = 10$ at $\Delta t = 0.01$. Euler drifts monotonically to 10.5%, a direct consequence of its non-symplectic character. Velocity-Verlet exhibits bounded oscillation at amplitude $\sim 2.5 \times 10^{-5}$ (visible in the inset), consistent with conservation of a shadow Hamiltonian. RK4's deviation is negligible ($1.4 \times 10^{-9}\%$) on this short interval, but its lack of symplecticity means energy would drift on longer timescales. This directly demonstrates the distinction between short-time local accuracy and long-time preservation of the invariant structure of the dynamics.

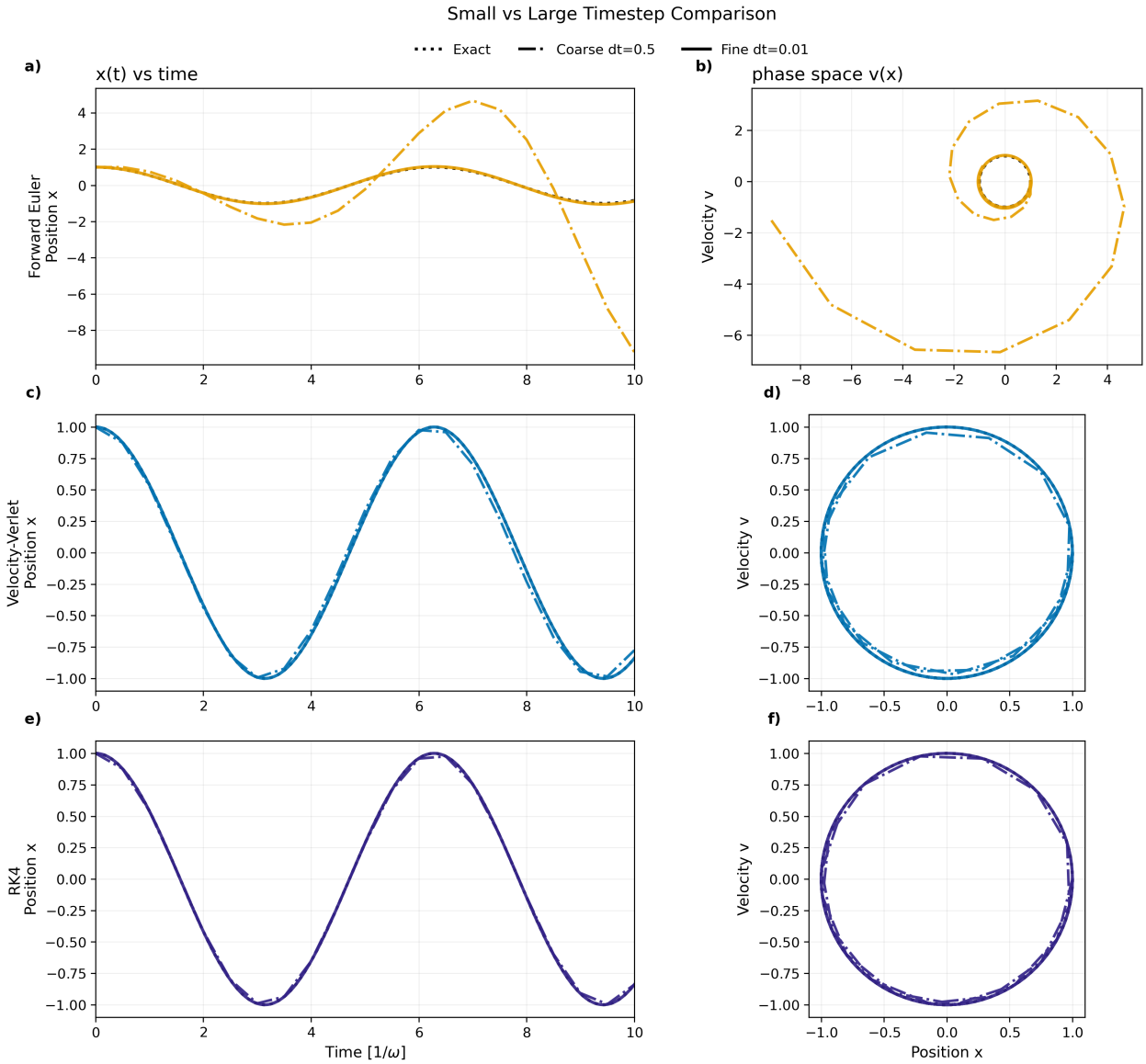

Figure 2. Comparison of coarse ($\Delta t = 0.5$, dash-dot) and fine ($\Delta t = 0.01$, solid) timesteps. Left: $x(t)$; right: phase-space portrait. Euler (orange, top) is unstable at $\Delta t = 0.5$; Velocity-Verlet (blue, centre) retains bounded orbits despite coarse resolution; RK4 (dark blue, bottom) remains close to exact at both resolutions.

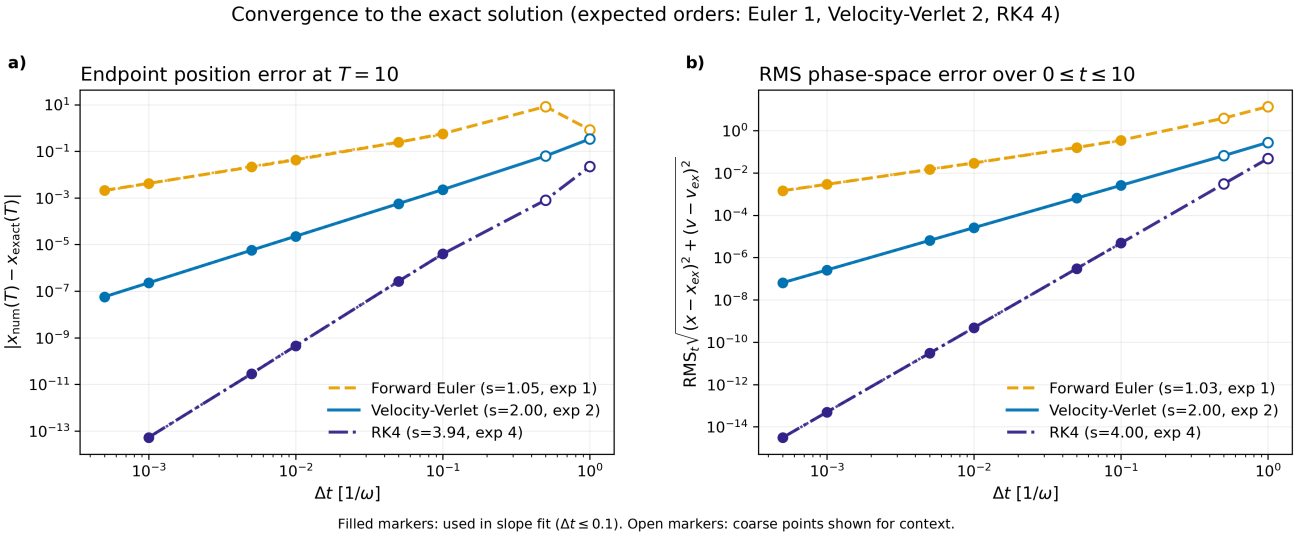

Figure 3. Convergence to the exact harmonic-oscillator solution ($\omega = 1$, $x_0 = 1$, $T = 10$). **(a)** Endpoint position error; **(b)** RMS phase-space error. Filled markers: asymptotic fit domain ($\Delta t \leq 0.1$); open markers: context only. Fitted slopes (Table 1) confirm first-, second-, and fourth-order convergence for Euler, Velocity-Verlet, and RK4. RK4 saturates at $\approx 3.6 \times 10^{-15}$ (IEEE 754 floor).

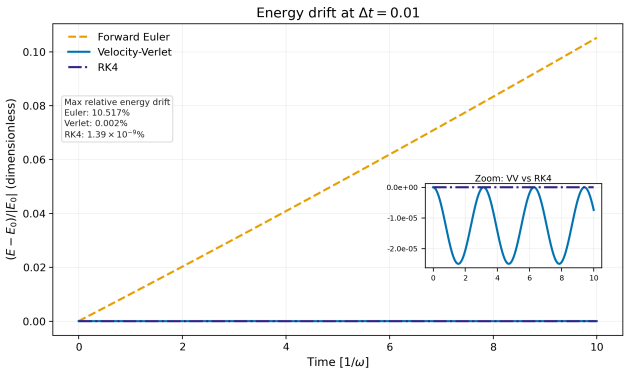

Figure 4. Relative energy deviation $(E - E_0)/|E_0|$ for the harmonic oscillator ($\Delta t = 0.01$, $T = 10$). Euler (orange) drifts monotonically to 10.5%. The inset reveals Velocity-Verlet (blue) bounded oscillation at $\sim 2.5 \times 10^{-5}$ (shadow-Hamiltonian conservation) and negligible RK4 deviation (1.4×10^{-9}%).

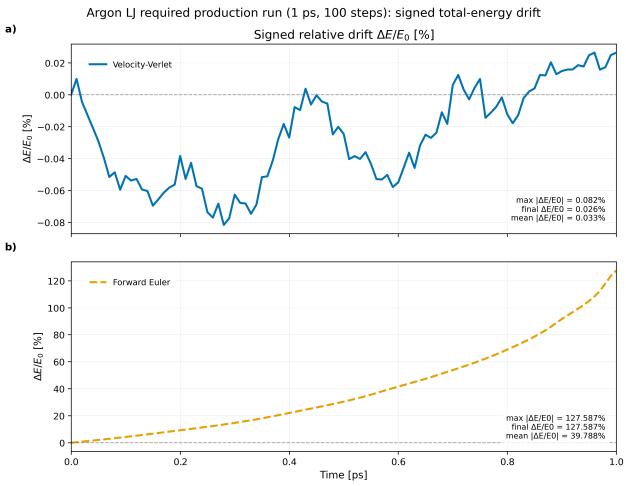

Figure 5. Signed relative energy deviation $\Delta E/E_0$ [%] during the 100-step NVE production run for liquid argon ($N = 864$, $\Delta t = 10^{-14}$ s, equilibrated to $T = 94.4$ K). **(a)** Velocity-Verlet: bounded oscillation, $\max|\Delta E/E_0| = 0.082\%$ (mean 0.033%), confirming symplectic long-time stability. **(b)** Forward Euler: monotonic drift to +127.6%, demonstrating catastrophic non-symplectic energy injection.

4 Results 2: Lennard-Jones Argon Fluid

The LJ argon system ($N = 864$, $\Delta t = 10^{-14}$ s) is simulated at $T = 94.4$ K using the equilibration protocol described in §2.3, followed by a clean 100-step NVE production run at $P = 4$.

Energy conservation: mandated 100-step run

Figure 5 shows the signed relative total-energy deviation $\Delta E/E_0$ for both integrators over the production run. Velocity-Verlet (panel a) maintains bounded, oscillatory deviation with $\max|\Delta E/E_0| = 0.082\%$ and mean 0.033%, consistent with shadow-Hamiltonian conservation (cf. Fig. 4). Forward Euler (panel b) diverges catastrophically to +127.6% within 1 ps, rendering the trajectory completely unphysical.

Three factors contribute to the finite Velocity-Verlet deviations. The hard cutoff at r_c introduces discontinuities whenever a particle pair crosses the cutoff radius, causing local energy spikes. The initial FCC lattice is far from the equilibrium liquid structure, generating large early-time forces before relaxation. Discrete-timestep truncation accumulates even in a symplectic scheme. A shifted-force cutoff—standard in production codes—would eliminate the force discontinuity at r_c.

Exact reproduction of Rahman's energy trajectory is not expected: the assignment brief explicitly notes sensitivity to initial conditions, and our FCC starting configuration differs from Rahman's prepared liquid state. Parallel correctness is confirmed by bitwise agreement of energy trajectories between $P = 1$ and $P = 2$ runs.

Temperature

Figure 6 confirms the temperature evolution. Velocity-Verlet holds $T = (94.4 \pm 1.4)$ K throughout, consistent with NVE energy conservation. Euler heats from the target 94.4 K to ∼400 K within 1 ps, a direct consequence of non-symplectic energy injection; mean temperature rises to 185 K over the 100 steps. This temperature runaway demonstrates conclusively that Euler is unsuitable for production MD even from a well-equilibrated initial configuration.

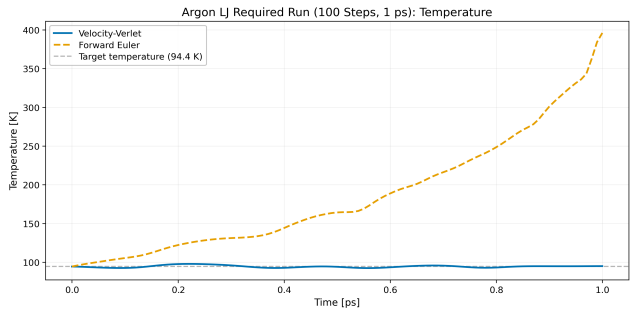

Figure 6. Instantaneous temperature during the 100-step argon NVE run ($N = 864$, $\Delta t = 10^{-14}$ s). Velocity-Verlet (blue) holds $T \approx 94.4$ K (grey dashed); Forward Euler (orange) heats to ∼400 K, reflecting the energy drift of Fig. 5.

Radial distribution function

To validate the structural properties of the simulated liquid state, an extended NVE run was performed (20 000 production steps, Velocity-Verlet, $P = 4$, $N = 864$, with a 200-step equilibration discard before $g(r)$ accumulation sampled every 5 steps). Figure 7 shows $g(r)$ alongside a reference curve derived from Rahman 7 Fig. 2 using WebPlotDigitizer, with paper-anchored points at identifiable features.

The first peak at $r/\sigma = 1.09$ ($g = 2.84$), the first minimum at $r/\sigma = 1.55$, the second peak at $r/\sigma = 2.07$ ($g = 1.25$), and the long-range tail ($\langle g \rangle = 1.002$ for $r > 4\sigma$) all compare favourably with Rahman's published liquid-structure data. The comparison is qualitative to semi-quantitative: exact agreement is not expected given different initial configurations and finite run length. The Rahman reference was constructed from manually anchored digitisation points and should not be over-interpreted.

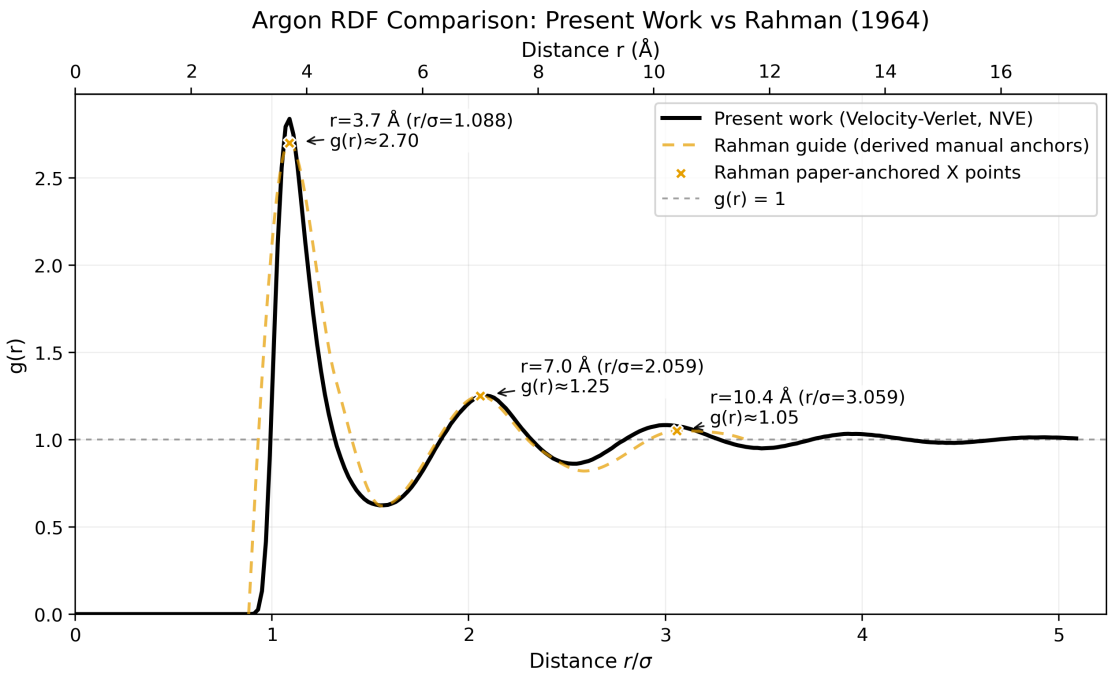

Figure 7. Radial distribution function $g(r)$ from an extended 20 000-step NVE run (Velocity-Verlet, $N = 864$, $P = 4$; $g(r)$ sampled every 5 steps after a 200-step discard), compared with the reference from Rahman 7 (orange dashed, digitised). First peak: $r/\sigma = 1.09$, $g = 2.84$; second peak: $r/\sigma = 2.07$, $g = 1.25$; tail $\langle g \rangle = 1.002 \to 1$. The comparison is qualitative to semi-quantitative; exact reproduction is not expected.

Table 2. Strong scaling results ($N = 2048$, Velocity-Verlet, 100 steps).

P	Wall [s]	Comm [s]	Comm%	$S(P)$	$E(P)$
1	45.96	0.001	0.0	1.00	1.000
2	23.15	0.244	1.1	1.99	0.993
4	12.00	0.661	5.5	3.83	0.957
8	6.09	0.554	9.1	7.55	0.944
16	3.21	0.367	11.4	14.30	0.894
24	2.37	0.319	13.5	19.42	0.809
32	1.91	0.290	15.2	24.12	0.754

5 Results 3: Parallel Scaling Analysis

All timing runs used Velocity-Verlet, fixed-seed initial conditions, $\Delta t = 10^{-14}$ s, and no file output, in accordance with the assignment brief.

Strong scaling

Figure 8a–c shows speedup $S(P) = T_1/T_P$, parallel efficiency $E(P) = S(P)/P$, and the computation/communication breakdown at fixed $N = 2048$. Table 2 records the measured values.

Scaling is near-ideal up to $P = 8$ (efficiency $> 94\%$). At $P = 32$, $S(32) = 24.1\times$ with efficiency 75%. Amdahl's law,

$$S(P) = \frac{1}{f + (1-f)/P}, \tag{10}$$

fitted via nonlinear least-squares across all $P > 1$ values yields $f \approx 0.010$. This is an *empirical descriptor* of the measured strong-scaling curve and should not be interpreted as a literal, machine-independent serial fraction: Amdahl's model assumes constant f and negligible communication, whereas the `MPI_Allgatherv` cost grows monotonically with P (Table 2, rising from 1.1% at $P = 2$ to 15.2% at $P = 32$), explaining the visible deviation from the Amdahl fit at high P. This is the expected behaviour of strong scaling: as the work per process falls, the communication term grows as a fraction of total runtime. A more realistic extension includes a message-passing overhead, $T_n = T_s + T_p/N + c\lambda N$, where λ represents the cost of inter-process communication and the final term captures the increasing impact of communication as process count rises [3].

Size scaling

Figure 9 shows wall time and its components as a function of N at fixed $P = 16$, ranging from $N = 108$ to $N = 2048$. Empirical log–log fits give wall time $\propto N^{1.85}$ and compute-only $\propto N^{1.93}$, close to the expected $\mathcal{O}(N^2)$ for direct all-pairs force evaluation. These are fits over a limited range of N and should be treated as such. The sub-quadratic wall-time exponent reflects the $\mathcal{O}(N)$ `MPI_Allgatherv` communication, which contributes a sub-quadratic component; the compute-only exponent converges toward 2 as N grows, confirming the $\mathcal{O}(N^2)$ force loop asymptotically dominates. Communication fraction falls monotonically from 51% at $N = 108$ to 13% at $N = 2048$ (Fig. 9b), confirming that $\mathcal{O}(N^2)$ force cost grows faster than $\mathcal{O}(N)$ communication volume.

Although only strong and size scaling are required here, it is helpful to note the complementary weak-scaling viewpoint: if work per process is held roughly constant while both problem size and process count increase together, one expects communication effects to be isolated more clearly, rather than being amplified by shrinking local workload.

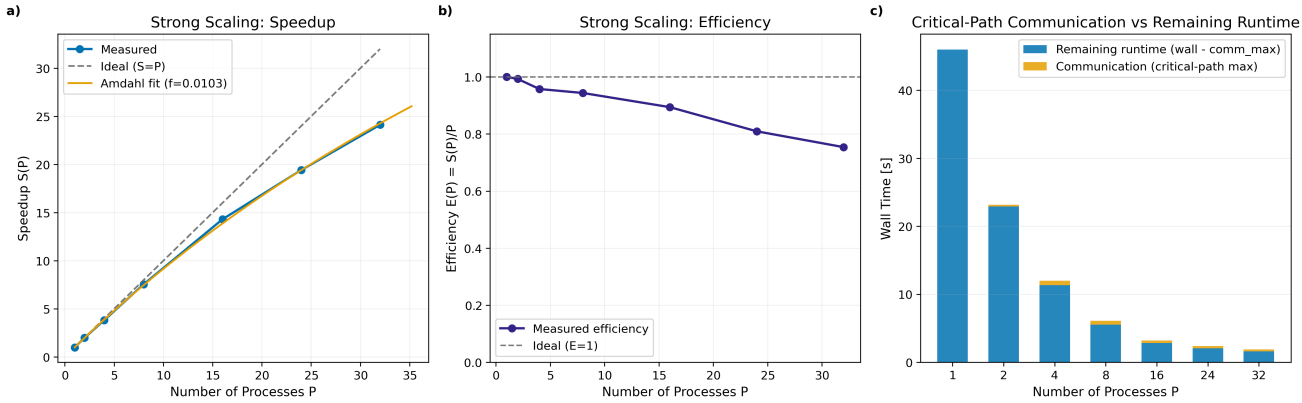

Figure 8. Strong scaling ($N = 2048$ fixed, Velocity-Verlet, 100 steps). **(a)** Speedup $S(P)$: ideal line (grey dashed), Amdahl fit (Eq. 10, $f = 0.010$, orange). Near-ideal to $P = 8$; $S(32) = 24.1\times$. **(b)** Parallel efficiency $E(P)$: above 94% for $P \leq 8$, 75% at $P = 32$. **(c)** Critical-path communication (orange) vs. remaining runtime (blue); communication fraction grows from 0% to 15%, explaining the Amdahl fit deviation at high P.

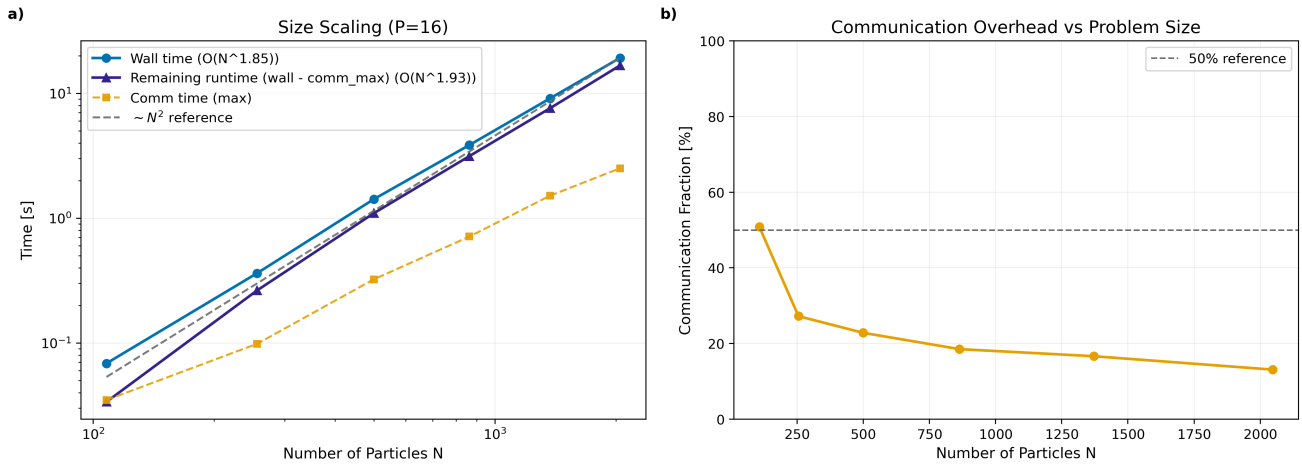

Figure 9. Size scaling ($P = 16$ fixed, Velocity-Verlet, 100 steps). **(a)** Log-log wall time (blue circles), compute-only (blue triangles), and communication time (orange squares) vs. N, with N^2 reference. Empirical fits: $T_{\text{wall}} \propto N^{1.85}$, $T_{\text{compute}} \propto N^{1.93}$. **(b)** Communication fraction vs. N: monotonic decrease from 51% to 13% confirms $\mathcal{O}(N^2)$ force cost grows faster than $\mathcal{O}(N)$ `MPI_Allgatherv` volume.

6 Conclusions

The harmonic oscillator tests confirm the expected convergence orders (1.05, 2.00, 3.94 for Euler, Velocity-Verlet, and RK4) and the decisive role of symplecticity in long-time energy conservation. Velocity-Verlet's bounded energy oscillation (0.002%) versus Euler's monotonic 10.5% drift shows that preserving the underlying Hamiltonian structure matters more for long-time MD stability than formal local order alone.

For the LJ argon fluid, Velocity-Verlet conserves energy to 0.082% over the mandated 100-step NVE run, while Euler diverges to 128% with temperature runaway to $\sim 400\,\text{K}$. This demonstrates conclusively that Euler is unsuitable for production MD. RK4 achieves the highest formal accuracy (down to the IEEE 754 floor at 3.6×10^{-15}) but requires four force evaluations and four `MPI_Allgatherv` calls per step; it is also non-symplectic and would drift on long timescales. Velocity-Verlet, requiring one force evaluation and one collective per step, is the correct choice.

Strong scaling achieves $24.1\times$ speedup at $P = 32$ with 75% efficiency (empirical Amdahl $f \approx 0.010$), limited at high P by growing `MPI_Allgatherv` overhead not captured by the simple Amdahl model. The fundamental bottleneck is $\mathcal{O}(N)$ communication volume per step: the full position array must be gathered every timestep. In distributed-memory computing, this cost is ultimately controlled by latency and bandwidth as well as algorithmic structure, so parallel performance is never determined by flop-count alone.

Production codes such as LAMMPS [9] and GROMACS use *spatial domain decomposition* with *neighbour lists* (Verlet or cell lists) to reduce force evaluation to $\mathcal{O}(N)$ and limit communication to $\mathcal{O}(N^{2/3}/P)$ halo exchange—a fundamental scaling advantage. Additional improvements include shifted-force cutoffs to eliminate energy discontinuities at r_c, and Ewald summation for long-range electrostatics. The complementary perspective of Gustafson–Barsis's law is relevant here: the size-scaling results show the code can tackle larger problems on more cores, even if the strong-scaling efficiency is not perfect [5].

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